

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 40)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

Table 1

Location	Mean wind speed / m s^{-1}	Standard deviation / m s^{-1}
Woodland	1.2	0.4
Clearing	3.5	1.1

0 1

. 1

Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

[3 marks]

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0 1

. 2

Explain what the student should do if it is not possible to show that the data are normally distributed.

[1 mark]

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0 1

. 3

The test gave $p = 0.02$. Explain what this means.

[2 marks]

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0 1

. 4

Explain how the standard deviations in Table 1 support the conclusion that the means are different.

[2 marks]

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0 1 . 5

Explain why a statistically significant result does not prove that trees cause the lower wind speed.

[2 marks]

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10

0 2

Figure 1 compares the energy flows in a conventional gas-fired power station and a combined heat and power (CHP) station, each supplied with 100 units of fuel energy.

Figure 1

	Conventional power station	CHP station
Electricity output / units	40	35
Useful heat supplied to buildings / units	0	45
Waste heat lost / units	60	20

0 2 . 1

Calculate the overall efficiency of each station.

[2 marks]

Answer %

0 2 . 2

Explain why the electrical efficiency of a CHP station is often deliberately kept below 40%.

[2 marks]

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0 2 . 3

Explain why CHP stations must be located close to the users of the heat.

[1 mark]

0 2 . 4

Explain how each of the following features of an industrial heat exchanger increases its efficiency.

[3 marks]

Long narrow copper pipes

Counter-current flow

Reduced flow rate

0 2 . 5

Explain why a single large spherical tank loses less heat than several small cubic tanks holding the same total volume of hot fluid.

[2 marks]

10

0 3 . 1

Distinguish between **renewable** and **depletable** energy resources, giving an example of a resource that is both renewable and depletable.

[3 marks]

03 . 2

Explain why the abundance of an energy resource is not the same as the amount available for use. Give **two** examples.

[3 marks]

03 . 3

Describe the locational constraints of **two** named renewable energy resources.

[4 marks]

10

0 4

A factory discharges a persistent, water-soluble pollutant into a river. Table 1 shows the concentration of the pollutant measured at sites along the river on a day with normal river flow.

Table 1

Site	Distance downstream of discharge / km	Pollutant concentration / mg dm ⁻³	River flow / m ³ s ⁻¹
A (upstream)	-2	0.0	12
B	0.5	4.8	12
C	5	3.9	15
D (below tributary)	12	1.6	36
E	30	1.4	38

0 4**1**

Calculate the percentage decrease in pollutant concentration between site B and site E.

[2 marks]

Answer

%

0 4**2**

Explain the large decrease in concentration between sites C and D.

[2 marks]

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0 4**3**

Explain why the concentration changes very little between sites D and E.

[2 marks]

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0 4**4**

Suggest why the pollution might be more serious if the same discharge entered a lake instead of a river.

[2 marks]

0 4 . 5

Explain why the factory might be permitted to discharge the effluent into the sea near a strong current but not into the river.

[2 marks]

10

0 5 . 1

Explain the locational requirements for harnessing wave power.

[2 marks]

0 5 . 2

Describe how **one** named wave power device generates electricity.

[2 marks]

0 5 . 3

Give **two** design problems that have delayed the development of wave power.

[2 marks]

1

2

05 . 4

Compare the environmental impacts of tidal barrages, tidal lagoons and in-stream tidal turbines.

[4 marks]

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06

Figure 1 shows a simplified energy budget for the Earth and its atmosphere. Values are percentages of incoming solar radiation.

Figure 1

Energy flow	Percentage of incoming solar radiation / %
Reflected by clouds, dust and atmosphere	25
Reflected by the Earth's surface	5
Absorbed by the atmosphere	20
Absorbed by the Earth's surface	50
Infrared emitted by the surface and absorbed by greenhouse gases	95
Infrared re-emitted by the atmosphere back to the surface	85
Infrared emitted to space	70

06 . 1

Use Figure 1 to calculate the planetary albedo of the Earth.

[1 mark]

Answer %

0 6 . 2 Describe how each of the following wavelengths of electromagnetic radiation behaves in the atmosphere. **[6 marks]**

Ultraviolet (UV)
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Visible light
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Infrared (IR)
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0 6 . 3 Explain why the amount of infrared re-emitted to the surface (85%) is larger than the amount emitted to space (70%). **[2 marks]**

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0 6 . 4 Explain the term **thermal stratification** of the atmosphere. **[1 mark]**

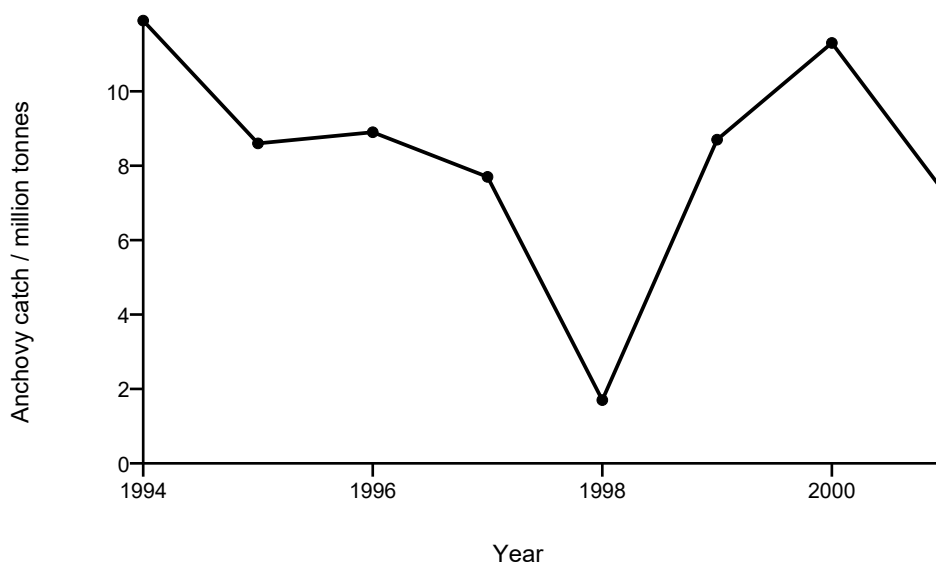
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10

07

Figure 1 shows the sea surface temperature anomaly in the eastern Pacific off Peru and the annual Peruvian anchovy catch for a series of years including a strong El Niño event in 1997-98.

Figure 1



07

. 1

Describe the normal (non-El Niño) pattern of winds and ocean currents in the tropical Pacific and explain why the Peruvian coast supports a rich fishery.

[4 marks]

07

. 2

Explain how an El Niño event caused the collapse of the anchovy catch in 1998 shown in Figure 1.

[3 marks]

0 7 . 3 Describe **two** other effects of El Niño events.

[2 marks]

1

2

0 7 . 4 Suggest why some scientists link global climate change to changes in the frequency of El Niño events, and why this link is uncertain.

[1 mark]

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10

0 8 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

- A Aerobic decomposition produces little methane and returns nutrients ☒ to the soil
- B It produces more electricity ☐
- C It removes heavy metals ☐
- D It destroys asbestos fibres ☐

0 8 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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08 . 3

Give **two** features of a well-managed landfill site that reduce pollution.**[2 marks]**

1

2

5

09

Table 1 shows global reactive nitrogen fixed each year by natural processes and by human activities.

Table 1

Source	Nitrogen fixed / million tonnes per year
Biological fixation (natural ecosystems)	110
Lightning	5
Haber process (fertiliser manufacture)	120
Cultivation of legume crops	60
Combustion (NO _x)	30

09 . 1

Use Table 1 to calculate the percentage of total nitrogen fixation that is caused by human activities.

[2 marks]

Answer

%

09 . 2

Describe the Haber process and explain why it is unsustainable compared with biological nitrogen fixation.

[3 marks]

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09 . 3

Describe **three** consequences of the increased movement of nitrogen compounds caused by human activities.

[3 marks]

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09 . 4

Explain how composting or anaerobic digestion of biological wastes produces a better fertiliser than raw waste.

[2 marks]

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10 . 1

Explain how the properties of smoke and of the gases involved in acid rain lead to their environmental effects, and evaluate the effectiveness of the methods used to control them.

[9 marks]

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Extra space

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10 . **2** State **one** effect of tropospheric ozone on plants.

[1 mark]

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

1	1	.	2
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Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS